The criterion for two lines to be parallel

One pair of equal alternate angles is enough — parallelism becomes something you can check.

LESSON 82–83 | 2 × 40 MIN | MATEMATIKA 7, §38 | S8 5.1

LESSON CLOCK

15' | 25' | 25' | 20' | 5'

The transversal and its eight angles	15 min
The three criteria	25 min
Using a criterion	25 min
Perpendiculars to one line	20 min
Homework	5 min

LEARNING OBJECTIVES

- Name the alternate, corresponding and co-interior angles at a transversal.
- State and use the three criteria for two lines to be parallel.
- Prove that two lines are parallel from an angle equality.
- Deduce that two perpendiculars to the same line are parallel.

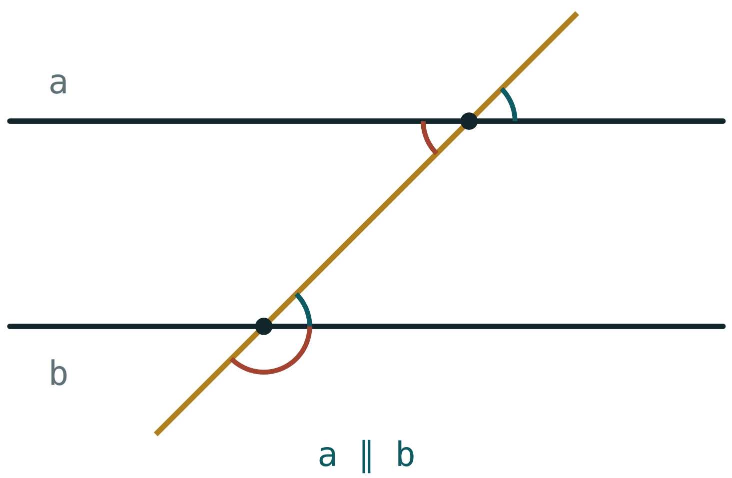
TEXTBOOK

National	pp. 231–237
Cambridge	Stage 8, pp. 44–47
Workbook	Exercise 5.1

01 Explanation

The transversal and its eight angles

A line that cuts two others is a **transversal**. It makes eight angles, and three pairs of them have names.



The transversal and the named pairs

PAIR	WHERE THEY SIT	WHEN THE LINES ARE PARALLEL
alternate	between the lines, opposite sides of the transversal	equal
corresponding	same side, one above and one below	equal
co-interior	between the lines, same side	add to $180°$

NAMES FIRST, FACTS SECOND

Half the errors in this chapter are naming errors, not reasoning errors. Point at a pair and say its name aloud before quoting any rule about it.

The three criteria

Each pair gives a criterion. Every one of them is a one-way street read this way: the angle fact comes first, and parallelism is the conclusion.

$$\text{alternate equal} \Longrightarrow a \parallel b \ \text{corresponding equal} \Longrightarrow a \parallel b \ \text{co-interior sum } 180° \Longrightarrow a \parallel b$$

GIVEN	CONCLUSION
$\angle 1 = \angle 2$	alternate — the lines are parallel
$\angle 1 = \angle 3$	corresponding — the lines are parallel
$\angle 1 + \angle 4 = 180°$	co-interior — the lines are parallel

ONE CORRECT PAIR IS ENOUGH — BUT IT MUST BE A NAMED PAIR

Two angles that happen to be equal prove nothing unless they are alternate or corresponding. Check the position first; the equality alone is not a criterion.

Using a criterion

The written proof is three lines: name the pair, quote the criterion, state the conclusion.

STATEMENT	REASON
$\angle 1 = 70°, \angle 2 = 70°$	given
they are alternate angles	position at the transversal
$a \parallel b$	the alternate-angle criterion

DATA	PAIR	PARALLEL?
$\angle 1 = 70°, \angle 2 = 70°$	alternate	yes
$\angle 1 = 110°, \angle 4 = 70°$	co-interior	yes
$\angle 1 = 70°, \angle 4 = 80°$	co-interior	no
$\angle 1 = 70°, \angle 3 = 70°$	corresponding	yes

Perpendiculars to one line

The neatest consequence: if two lines are both perpendicular to a third, they are parallel.

$$a \perp c \text{ and } b \perp c \Longrightarrow a \parallel b$$

STATEMENT	REASON
$a \perp c, b \perp c$	given
the two angles at c are $90°$ each	definition of perpendicular
they are corresponding and equal	position at the transversal
$a \parallel b$	the corresponding-angle criterion

THIS IS WHAT A SET SQUARE DOES

Sliding a set square along a ruler draws parallel lines because every line it draws is perpendicular to the ruler. The tool is the theorem made of plastic.

02 Worked examples

EXAMPLE 1 **A transversal makes $\angle 1 = 64°$ and its alternate $\angle 2 = 64°$. Are the lines parallel?**

1. The pair is alternate — check the position.
2. They are equal.
3. By the alternate-angle criterion, $a \parallel b$.

ANSWER

Yes

EXAMPLE 2 **Co-interior angles measure $115°$ and $65°$. Are the lines parallel?**

1. $115° + 65° = 180°$
2. Co-interior angles adding to $180°$ is the criterion.
3. So the lines are parallel.

ANSWER

Yes

EXAMPLE 3 **Prove that $a \perp c$ and $b \perp c$ give $a \parallel b$.**

1. Both lines meet c at $90°$.
2. Those two right angles are corresponding.
3. They are equal.
4. By the corresponding-angle criterion, $a \parallel b$.

ANSWER

$a \parallel b$

03 Interactive model

Draw one transversal on the board and have the class colour the three named pairs in three colours before any rule is stated; the naming errors disappear.

Name the pair, then decide

Angles between the lines, opposite sides of the transversal:

- corresponding
- alternate
- co-interior
- vertical

Angles on the same side, one above and one below:

- corresponding
- alternate
- co-interior
- adjacent

Angles between the lines on the same side:

- corresponding
- alternate
- co-interior
- vertical

Equal alternate angles give:

- perpendicular lines
- parallel lines
- nothing
- a triangle

Co-interior angles that add to $180°$ give:

- parallel lines
- intersecting lines
- nothing
- equal lines

$\angle 1 = 70°$ and co-interior $\angle 4 = 80°$:

- parallel
- not parallel
- perpendicular
- cannot say

Two lines perpendicular to a third are:

- perpendicular
- parallel
- skew
- equal

Two equal angles that are not a named pair prove:

- parallelism
- nothing
- perpendicularity
- congruence

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Transversal	Kesuvchi	Секущая
Alternate angles	Ichki almashinuvchi burchaklar	Накрест лежащие углы
Corresponding angles	Mos burchaklar	Соответственные углы
Co-interior angles	Ichki bir tomonli burchaklar	Односторонние углы
Criterion	Alomat	Признак
To conclude	Xulosa chiqarmoq	Сделать вывод
Perpendicular	Perpendikulyar	Перпендикуляр
Proof	Isbot	Доказательство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A line cutting two others is a:

- bisector
- transversal
- median
- chord

A transversal makes how many angles?

- 4
- 6
- 8
- 12

Alternate angles equal $\Longrightarrow$

- the lines meet
- the lines are parallel
- nothing
- a right angle

Co-interior angles of parallel lines add to:

- $90°$
- $180°$
- $270°$
- $360°$

Corresponding angles of $64°$ and $64°$ give:

- parallel lines
- perpendicular lines
- nothing
- a triangle

$a \perp c$ and $b \perp c$ give:

- $a \perp b$
- $a \parallel b$
- $a = b$
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Name the pair between the lines, opposite sides

   ANSWER alternate

2. Name the pair on the same side, one above one below

   ANSWER corresponding

3. Name the pair between the lines, same side

   ANSWER co-interior

4. Alternate angles $70°$ and $70°$: parallel?

   ANSWER Yes

5. Corresponding angles $50°$ and $60°$: parallel?

   ANSWER No

6. Co-interior $100°$ and $80°$: parallel?

   ANSWER Yes

7. A transversal makes how many angles?

   ANSWER 8

Medium · the standard question

7 PROBLEMS

1. Co-interior $115°$ and $65°$

   ANSWER Parallel

2. Co-interior $70°$ and $80°$

   ANSWER Not parallel

3. Alternate $3x$ and $75°$ with the lines parallel: find x

   ANSWER $x = 25$

4. Corresponding $2x + 10$ and $80°$ with the lines parallel

   ANSWER $x = 35$

5. $a \perp c$, $b \perp c$: what follows?

   ANSWER $a \parallel b$

6. Write the three-line proof for equal alternate angles

   ANSWER given; alternate; criterion

7. Two equal angles that are vertical prove parallelism?

   ANSWER No — not a named pair

Hard · stretch and reason

7 PROBLEMS

1. Co-interior angles are x and $2x$ with the lines parallel: find both

   ANSWER $60°$ and $120°$

2. Alternate angles are $4x - 10$ and $3x + 20$: find x for parallelism

   ANSWER $x = 30$

3. Co-interior $5x$ and $4x$: find x

   ANSWER $x = 20$

4. Why is each criterion stated as an implication, not an equality?

   ANSWER The angle fact is the evidence; parallelism is the conclusion

5. A set square slid along a ruler draws parallels — why?

   ANSWER Every line drawn is perpendicular to the ruler

6. Corresponding angles $x + 30$ and $2x - 10$: find x

   ANSWER $x = 40$

7. Given $\angle 1 = 108°$, find the co-interior angle that makes the lines parallel

   ANSWER $72°$

07 Homework

Homework — 5 tasks

Name the pair in words before you write a single number.

1. Draw a transversal across two lines and label all three named pairs.
2. Alternate angles are $5x$ and $65°$. Find x if the lines are parallel.
3. Co-interior angles are $3x$ and $2x$. Find both.
4. Write the three-line proof that $a \perp c$ and $b \perp c$ give $a \parallel b$.
5. Explain why two equal vertical angles do not prove that two lines are parallel.